

# Rahul Bouri

 raoulbouri |  raoul-bouri |  rahulbouri16@gmail.com |  +919711602325

## Summary

ML engineer building AI systems at the intersection of **robotics**, **foundation models**, and **production ML**. Experienced in multimodal perception, agentic systems, and scalable ML infrastructure, with interests in **Physical AI**, **world models**, **reinforcement learning**, and **learning-based control**.

## Education

|             |                                                                                                                                                                                                                                                                                                                                                       |                    |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2026 – 2028 | MS Robotic Systems Development at <b>Carnegie Mellon University (CMU)</b><br><b>Coursework:</b> Robo-Learning, Deep Learning, Advanced Computer Vision                                                                                                                                                                                                | GPA: 4/4 (ongoing) |
| 2020 – 2024 | BE Mechanical Engineering at <b>Birla Institute of Technology and Science (BITS), Pilani</b><br><b>Coursework:</b> Linear Algebra, Multivariable Calculus, Control Systems, Kinematics and Dynamics of Machines, Computer Aided Design (CAD)<br><b>AI Minor Coursework:</b> Machine Learning, Deep Learning, Applied Statistical Methods, Data Mining | CGPA: 8.6/10       |

## Work Experience

**AI Research Engineer, AlphaSense** *Bangalore, India* *December 2025–July 2026*

- **Citation Service for Generative Search** *Microservice grounding LLM responses to enterprise source document*
  - Improved citation coverage by combining lexical methods with LLM-based extraction, reserving model inference for low-confidence cases to balance accuracy, latency and cost.
  - Productionized the service for **100 RPS** at **p90 latency < 3s**, enabling citation-backed Generative Search responses.
- **Agent Registry and Custom Workflow Agents** *Infrastructure for reusable agent workflows through a centralized registry*
  - Established a centralized agent registry that standardized capability discovery, onboarding, and lifecycle management.
  - Enabled **100+ agents** to interoperate via a **standardized REST API**, serving over **30% of weekly search traffic**.

**Machine Learning Engineer, OLA Cabs** *Bangalore, India* *August 2024–November 2025*

- **Enterprise Agentic Query Engine** *Enabled leadership to monitor organization performance via natural language queries*
  - Built **fault-tolerant data pipelines** with checkpointing to continuously ingest operational metrics while minimizing cost.
  - Developed an **agentic analytics engine** that translated **natural-language queries** into validated SQL through semantic retrieval and structured error recovery.
  - Adopted by the **Founder’s Office** and business leaders for weekly operational reviews, achieving **92% task success** and **85% precision** on an enterprise benchmark.
- **Food Delivery Time Prediction** *Improving customer experience by reducing delivery delays and cancellations*
  - Built a streaming feature pipeline with continuous feature computation for real-time ETA prediction.
  - Enabled the service with automated monitoring and drift detection to ensure reliable low-latency deployment.
  - Achieved **95% ETA accuracy** across **100k+ daily orders**, reducing cancellations from **26% to 14%**.
- **OCR for Driver & Customer Onboarding** *Accelerated onboarding and reduced manual verification costs across India*
  - Developed a document processing pipeline that generalized across **15+ Indian government-issued documents** through vision-based segmentation and OCR.
  - Deployed the service to process **10k+ daily requests** with **98% accuracy**, enabling nationwide rollout and saving over **INR 5 lakh/month**.

## Internships

**Data Science Intern, PwC USA** *Bangalore, India* *January 2024–June 2024*

Developed an agentic retrieval framework combining **vector search**, a **knowledge graph**, and a fine-tuned **Llama-13B**, improving retrieval precision by **30%**, reducing token usage by **75%**, and achieving **71% task accuracy** on Indian tax documents.

**Robotics Intern, TCS Research** *New Delhi, India* *June 2023–December 2023*

Implemented RGB-D SLAM for reliable indoor navigation in an assistive robot. Enhanced the perception stack with multi-modal emotion recognition (**74% weighted accuracy**) and emotion-aware speech synthesis for natural human-robot interaction.

## Publications

Bouri, Rahul et al. (July 2025). “Agentic LLMs for Question Answering over Tabular Data”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Association for Computational Linguistics, pp. 2249–2255. isbn: 979-8-89176-273-2. url: <https://aclanthology.org/2025.semeval-1.292/>.

Bouri, Rahul et al. (Sept. 2025). *From Data to Alloys: Predicting and Screening High Entropy Alloys for High Hardness Using Machine Learning*. arXiv: 2509.13479 [cond-mat.mtrl-sci]. url: <https://arxiv.org/abs/2509.13479>.

Bouri, Rahul et al. (July 2025). “Improving Narrative Classification and Explanation via Fine Tuned Language Models”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Association for Computational Linguistics, pp. 2233–2239. isbn: 979-8-89176-273-2. url: <https://aclanthology.org/2025.semeval-1.290/>.

## Skills

|                   |                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------|
| Programming       | Python, C++, R, SQL, PySpark, ROS, PyTorch, Tensorflow, MLFlow, Ray, Uvicorn, MATLAB, Git, AWS, Fusion 360, $\LaTeX$ |
| Certifications    | UCSD: DSA, Discrete Math for CS, GANs, Multivariate Calculus for ML                                                  |
| Extra Curriculars | Football, Swimming, Scuba Diving(PADI Certified), Photography                                                        |